# ENGINEERING MATHEMATICS I (EM201T / NEM201T)

## MASTER QUESTION BANK & EXTRA PRACTICE ARCHIVE (v2)

**Includes 100% Original PYQs (Mid-Sems 2024–26, SEEs 2024–25, Re-Exam 2026, Assignments) + Topic-Wise Extra Practice Problems**

## Topic 1.1: Limits & Asymptotic Behavior

### ORIGINAL EXAM PYQ / ASSIGNMENT: Indeterminate Form $\left[\frac{0}{0}\right]$ & Limits at Infinity (Mid-Sem 2024, 2025, 2026, Assignment Q1, Q3, Q4, Q6, Q7)

**1. Evaluate** $\lim_{x\to 5} \frac{x-5}{x^2-25}$.
*Solution*: Factorize denominator: $x^2-25=(x-5)(x+5)$. Cancel common factor $(x-5)$:

$$\lim_{x\to 5} \frac{1}{x+5} = \frac{1}{5+5} = \mathbf{\frac{1}{10}} \quad (0.1)$$

**2. Evaluate** $\lim_{x\to 3} \frac{x^2-7x+12}{x-3}$.
*Solution*: Factorize numerator: $x^2-7x+12=(x-3)(x-4)$. Cancel $(x-3)$:

$$\lim_{x\to 3}(x-4) = 3-4 = \mathbf{-1}$$

**3. Evaluate** $\lim_{x\to\infty} \frac{x-5x^2}{x^2-22x+3x}$.
*Solution*: Simplify denominator to $x^2-19x$. Divide numerator and denominator by highest power $x^2$:

$$\lim_{x\to\infty} \frac{\frac{1}{x}-5}{1-\frac{19}{x}} = \frac{0-5}{1-0} = \mathbf{-5}$$

**4. Evaluate** $\lim_{x\to 2} \frac{x^{10}-2^{10}}{x-2}$.
*Solution*: Apply standard power limit formula $\lim_{x\to a} \frac{x^n-a^n}{x-a} = na^{n-1}$ with $a=2, n=10$:

$$\text{Limit} = 10\cdot 2^{10-1} = 10\cdot 2^9 = 10\cdot 512 = \mathbf{5120}$$

**5. Evaluate** $\lim_{x\to 0} \frac{\sin 5x}{4x}$ **and** $\lim_{x\to 0} \frac{\tan^2 5x}{x^2}$.
*Solution*: Using $\lim_{\theta\to 0} \frac{\sin\theta}{\theta}=1$ and $\lim_{\theta\to 0} \frac{\tan\theta}{\theta}=1$:

$$\lim_{x\to 0} \frac{5}{4}\left(\frac{\sin 5x}{5x}\right) = \mathbf{\frac{5}{4}}, \qquad \lim_{x\to 0}\left(\frac{\tan 5x}{x}\right)^2 = \lim_{x\to 0}\left(5\cdot\frac{\tan 5x}{5x}\right)^2 = 5^2(1)^2 = \mathbf{25}$$

### EXTRA PRACTICE QUESTION: Algebraic, Power & Trigonometric Limits

**Practice Q1. Evaluate** $\lim_{x\to 4} \frac{x^2-16}{x^2-3x-4}$.
*Solution*: Factor numerator $(x-4)(x+4)$ and denominator $(x-4)(x+1)$. Cancel factor $(x-4)$:

$$\lim_{x\to 4} \frac{x+4}{x+1} = \frac{4+4}{4+1} = \mathbf{\frac{8}{5}}$$

**Practice Q2. Evaluate** $\lim_{x\to 0} \frac{1-\cos 4x}{x^2}$.
*Solution*: Use double-angle trigonometric identity $1-\cos 4x = 2\sin^2 2x$:

$$\lim_{x\to 0} \frac{2\sin^2 2x}{x^2} = 2\cdot\lim_{x\to 0}\left(\frac{\sin 2x}{x}\right)^2 = 2\cdot\left(2\cdot\lim_{x\to 0}\frac{\sin 2x}{2x}\right)^2 = 2\cdot(2)^2 = \mathbf{8}$$

**Practice Q3. Evaluate** $\lim_{x\to\infty} \frac{4x^3-2x+7}{2x^3+5x^2-1}$.
*Solution*: Divide numerator and denominator by highest power $x^3$:

$$\lim_{x\to\infty} \frac{4-\frac{2}{x^2}+\frac{7}{x^3}}{2+\frac{5}{x}-\frac{1}{x^3}} = \frac{4-0+0}{2+0-0} = \mathbf{2}$$

## Topic 1.2: Piecewise Boundary Continuity (Parameter Solvers)

### ORIGINAL EXAM PYQ / ASSIGNMENT: Solving Boundary Parameters for Continuity (Mid-Sem 2024, 2025, Re-Exam 2026)

**1. Consider** $f(x) = \begin{cases} A + x & x < 2 \\ 1 + x^2 & x \geq 2 \end{cases}$**. Find** $A$ **for continuity at** $x = 2$**.**

*Solution*: Equate Left-Hand Limit ($\lim_{x \to 2^-} f(x)$) and Right-Hand Limit ($f(2)$):

$$\text{LHL} = A + 2, \qquad \text{RHL} = 1 + (2)^2 = 5 \implies A + 2 = 5 \implies \mathbf{A = 3}$$

**2. Consider** $f(x) = \begin{cases} 2x^2 + 3 & x \leq 2 \\ 3x + B & x > 2 \end{cases}$**. Find** $B$ **for continuity at** $x = 2$**.**

*Solution*: Equate $f(2)$ and RHL ($\lim_{x \to 2^+} f(x)$):

$$f(2) = 2(2)^2 + 3 = 11, \qquad \text{RHL} = 3(2) + B = 6 + B \implies 6 + B = 11 \implies \mathbf{B = 5}$$

### EXTRA PRACTICE QUESTION: Boundary Value Parameters in Piecewise Functions

**Practice Q1. Find** $k$ **such that** $f(x) = \begin{cases} kx^2 + 1 & x \leq 3 \\ 2x + 7 & x > 3 \end{cases}$ **is continuous at** $x = 3$**.**

*Solution*: Compute LHL and RHL at $x = 3$:

$$\text{LHL} = f(3) = k(3)^2 + 1 = 9k + 1, \qquad \text{RHL} = 2(3) + 7 = 13$$

Equating LHL = RHL: $9k + 1 = 13 \implies 9k = 12 \implies \mathbf{k = \frac{4}{3}}$

**Practice Q2. Find** $m$ **and** $c$ **such that** $f(x) = \begin{cases} mx + 2 & x < 1 \\ 5 & x = 1 \\ x^2 + c & x > 1 \end{cases}$ **is continuous at** $x = 1$**.**

*Solution*: For continuity at $x = 1$, we must have LHL = RHL = $f(1) = 5$:

$$\text{LHL} : m(1) + 2 = 5 \implies \mathbf{m = 3}, \qquad \text{RHL} : (1)^2 + c = 5 \implies \mathbf{c = 4}$$

## Topic 1.3: Partial Differentiation & Clairaut's Theorem

### ORIGINAL EXAM PYQ / ASSIGNMENT: First & Second-Order Partial Derivatives (Mid-Sem 2024–26, SEE 2024–25, Assignment Q24)

**1. Find all partial derivatives of $f(x,y) = x^2 + y^2 + x^2y^2$.**
*Solution*:

$$\frac{\partial f}{\partial x} = 2x + 2xy^2 = \mathbf{2x(1+y^2)}, \qquad \frac{\partial f}{\partial y} = 2y + 2x^2y = \mathbf{2y(1+x^2)}$$

**2. Verify $f_{xy} = f_{yx}$ for $u = x^2e^{5y}$.**
*Solution*: First compute pure partials: $\frac{\partial u}{\partial y} = 5x^2e^{5y}$ and $\frac{\partial u}{\partial x} = 2xe^{5y}$.

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x}(5x^2e^{5y}) = \mathbf{10xe^{5y}}, \qquad \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial}{\partial y}(2xe^{5y}) = \mathbf{10xe^{5y}} \quad \text{(Verified Equal)}$$

### EXTRA PRACTICE QUESTION: Logarithmic & Mixed Partial Derivatives

**Practice Q1. Find $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$ for $u = \ln(x^2+y^2)$.**
*Solution*: Apply chain rule for logarithms:

$$\frac{\partial u}{\partial x} = \frac{1}{x^2+y^2} \cdot \frac{\partial}{\partial x}(x^2+y^2) = \frac{\mathbf{2x}}{\mathbf{x^2+y^2}}, \qquad \frac{\partial u}{\partial y} = \frac{1}{x^2+y^2} \cdot \frac{\partial}{\partial y}(x^2+y^2) = \frac{\mathbf{2y}}{\mathbf{x^2+y^2}}$$

**Practice Q2. Verify Clairaut's mixed partial theorem $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$ for $u = x^3y^2 + \sin(xy)$.**
*Solution*:

$$\frac{\partial u}{\partial y} = 2x^3y + x\cos(xy) \implies \frac{\partial^2 u}{\partial x \partial y} = \mathbf{6x^2y} + \cos(\mathbf{xy}) - \mathbf{xy}\sin(\mathbf{xy})$$

$$\frac{\partial u}{\partial x} = 3x^2y^2 + y\cos(xy) \implies \frac{\partial^2 u}{\partial y \partial x} = \mathbf{6x^2y} + \cos(\mathbf{xy}) - \mathbf{xy}\sin(\mathbf{xy}) \quad \text{(Equal)}$$

## Topic 1.4: Multivariable Chain Rule & Total Differentials

### ORIGINAL EXAM PYQ / ASSIGNMENT: Total Differential Coefficient & Implicit Derivatives (Mid-Sem 2024, 2025, 2026, SEE 2024, 2025)

**1. Find total differential coefficient of $u = x^2y$ w.r.t $x$ where $x^2 + xy + y^2 = 1$.**
*Solution*: Total differential formula $\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\frac{dy}{dx}$. 1. $\frac{\partial u}{\partial x} = 2xy$, $\frac{\partial u}{\partial y} = x^2$. 2. Implicit differentiation of $x^2 + xy + y^2 = 1 \implies 2x + y + x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{2x+y}{x+2y}$. 3. Substitute: $\frac{du}{dx} = 2xy + x^2\left(-\frac{2x+y}{x+2y}\right) = \frac{2x^2y+4xy^2-2x^3-x^2y}{x+2y} = \frac{\mathbf{3x^2y^2-2x^3}}{\mathbf{x+2y}}$.

**2. Find $\frac{dy}{dx}$ implicitly if $x^3 + y^2 = 16$.**
*Solution*: Differentiate both sides w.r.t $x$: $3x^2 + 2y\frac{dy}{dx} = 0 \implies \mathbf{\frac{dy}{dx} = -\frac{3x^2}{2y}}$.

### EXTRA PRACTICE QUESTION: Implicit Differentiation & Polar Chain Rule

**Practice Q1. Find $\frac{dy}{dx}$ implicitly for $x^3 + y^3 - 3axy = 0$ (Folium of Descartes).**
*Solution*: Differentiate term by term w.r.t $x$:

$$3x^2 + 3y^2\frac{dy}{dx} - 3a\left(y + x\frac{dy}{dx}\right) = 0 \implies (3y^2 - 3ax)\frac{dy}{dx} = 3ay - 3x^2 \implies \mathbf{\frac{dy}{dx} = \frac{ay - x^2}{y^2 - ax}}$$

**Practice Q2. Let $z = x^2 + y^2$ where $x = r\cos\theta$ and $y = r\sin\theta$. Find $\frac{\partial z}{\partial r}$ and $\frac{\partial z}{\partial \theta}$.**
*Solution*: Using chain rule: $\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial r} = (2x)(\cos\theta) + (2y)(\sin\theta) = 2r\cos^2\theta + 2r\sin^2\theta = \mathbf{2r}$.

$$\frac{\partial z}{\partial \theta} = (2x)(-r\sin\theta) + (2y)(r\cos\theta) = -2r^2\sin\theta\cos\theta + 2r^2\sin\theta\cos\theta = \mathbf{0}$$

## Topic 1.5: Homogeneous Functions & Euler's Theorem

### ORIGINAL EXAM PYQ / ASSIGNMENT: Homogeneous Degree & Euler Identity Proofs (Mid-Sem 2025, 2026, Assignment Q27)

**1. Test homogeneity of $f(x,y) = x^3 + 3x^2y + 3xy^2 + y^3$ and prove $xu_x + yu_y = 3u$.**
*Solution*: $f(tx,ty) = (tx)^3 + 3(tx)^2(ty) + 3(tx)(ty)^2 + (ty)^3 = t^3 f(x,y) \implies$ Homogeneous of degree $n = 3$. By Euler's Theorem $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu = \mathbf{3u}$.

### EXTRA PRACTICE QUESTION: Trigonometric Homogeneous Derivatives

**Practice Q1. If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \sin 2u$.**
*Solution*: Let $z = \tan u = \frac{x^3+y^3}{x-y}$. Degree of $z$ is $n = 3-1 = 2$. By Euler's theorem on $z$: $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 2z \implies x(\sec^2 u u_x) + y(\sec^2 u u_y) = 2\tan u$.

$$xu_x + yu_y = \frac{2\tan u}{\sec^2 u} = 2\sin u\cos u = \sin \mathbf{2u}$$

**Practice Q2. Determine the degree $n$ of the homogeneous function $f(x,y) = \frac{\sqrt{x}+\sqrt{y}}{x+y}$.**
*Solution*: $f(tx,ty) = \frac{\sqrt{tx}+\sqrt{ty}}{tx+ty} = \frac{t^{1/2}(\sqrt{x}+\sqrt{y})}{t(x+y)} = t^{-1/2} f(x,y) \implies \mathbf{n = -1/2}$.

## Topic 1.6: Directional Derivatives & Vector Gradient

### ORIGINAL EXAM PYQ / ASSIGNMENT: 3D Vector Gradient & Directional Derivatives (Mid-Sem 2025–26, SEE 2024–25, Re-Exam 2026)

**1. For $\Phi(x,y,z) = 3x^2y - y^3z^2$, find $\nabla\Phi$ at $(1,-2,-1)$.**
*Solution*: $\nabla\Phi = \frac{\partial\Phi}{\partial x}\mathbf{i} + \frac{\partial\Phi}{\partial y}\mathbf{j} + \frac{\partial\Phi}{\partial z}\mathbf{k} = (6xy)\mathbf{i} + (3x^2 - 3y^2z^2)\mathbf{j} - (2y^3z)\mathbf{k}$. At $(1,-2,-1)$: $\nabla\Phi = 6(1)(-2)\mathbf{i} + (3(1) - 3(4)(1))\mathbf{j} - (2(-8)(-1))\mathbf{k} = \mathbf{-12i - 9j - 16k}$.
**2. Find directional derivative of $f(x,y,z) = xyz$ at $(1,-1,1)$ in direction $3\mathbf{i} - 4\mathbf{k}$.**
*Solution*: 1. $\nabla f = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$. At $(1,-1,1) \implies \nabla f = -\mathbf{i} + \mathbf{j} - \mathbf{k}$. 2. Unit direction vector $\hat{\mathbf{u}} = \frac{3\mathbf{i}-4\mathbf{k}}{\sqrt{3^2+(-4)^2}} = \frac{3\mathbf{i}-4\mathbf{k}}{5}$.
3. $D_{\hat{\mathbf{u}}} f = \nabla f \cdot \hat{\mathbf{u}} = (-\mathbf{i} + \mathbf{j} - \mathbf{k}) \cdot \left(\frac{3\mathbf{i}-4\mathbf{k}}{5}\right) = \frac{(-1)(3)+(0)+(-1)(-4)}{5} = \frac{\mathbf{1}}{\mathbf{5}}$.

### EXTRA PRACTICE QUESTION: Maximum Rate of Change & 3D Directional Derivatives

**Practice Q1. Find the directional derivative of $f(x,y,z) = x^2yz + 4z^2$ at $(1,-2,-1)$ in direction $2\mathbf{i} - \mathbf{j} - 2\mathbf{k}$.**
*Solution*: $\nabla f = (2xyz)\mathbf{i} + (x^2z)\mathbf{j} + (x^2y + 8z)\mathbf{k}$. At $(1,-2,-1) \implies \nabla f = 4\mathbf{i} - \mathbf{j} - 10\mathbf{k}$. Unit vector $\hat{\mathbf{u}} = \frac{2\mathbf{i}-\mathbf{j}-2\mathbf{k}}{\sqrt{4+1+4}} = \frac{2\mathbf{i}-\mathbf{j}-2\mathbf{k}}{3}$.

$$D_{\hat{\mathbf{u}}} f = \nabla f \cdot \hat{\mathbf{u}} = \frac{(4)(2) + (-1)(-1) + (-10)(-2)}{3} = \frac{8+1+20}{3} = \mathbf{\frac{29}{3}}$$

**Practice Q2. Find the maximum rate of change of $\Phi(x,y,z) = x^2y^3z^4$ at $(1,1,1)$.**
*Solution*: $\nabla\Phi = (2xy^3z^4)\mathbf{i} + (3x^2y^2z^4)\mathbf{j} + (4x^2y^3z^3)\mathbf{k} = 2\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$ at $(1,1,1)$.

$$\text{Max Rate of Change} = |\nabla\Phi| = \sqrt{2^2 + 3^2 + 4^2} = \sqrt{4+9+16} = \sqrt{\mathbf{29}}$$

## Topic 1.7: Tangents, Normals & Relative Extrema

### ORIGINAL EXAM PYQ / ASSIGNMENT: Tangents, Normals & 2-Variable Extrema (SEE 2024–25, Re-Exam 2026)

**1. Find tangent and normal to $y = x + \frac{1}{x}$ at $x = 2$.**
*Solution*: Point: $x = 2 \implies y = 2 + 1/2 = 5/2 \implies (2, 5/2)$. Slope $m = \frac{dy}{dx} = 1 - \frac{1}{x^2} = 1 - 1/4 = 3/4$. Tangent: $y - 5/2 = 3/4(x - 2) \implies \mathbf{3x - 4y + 4 = 0}$. Normal: $y - 5/2 = -4/3(x - 2) \implies \mathbf{8x + 6y - 31 = 0}$.
**2. Classify extrema of $f(x, y) = 6x^2 + 10y^2 + 2x - 3y + 15$.**
*Solution*: 1. Critical points: $f_x = 12x + 2 = 0 \implies x = -1/6$; $f_y = 20y - 3 = 0 \implies y = 3/20$. Point: $(-1/6, 3/20)$. 2. Second partials: $A = f_{xx} = 12$, $B = f_{xy} = 0$, $C = f_{yy} = 20$. 3. Discriminant $\Delta = AC - B^2 = (12)(20) - 0 = 240 > 0$. Since $\Delta > 0$ and $A = 12 > 0$, it is a **Relative Minimum**.

### EXTRA PRACTICE QUESTION: 3D Tangent Planes & Saddle Points

**Practice Q1. Find the tangent plane to the surface $z = x^2 + y^2$ at $(1, 2, 5)$.**
*Solution*: $F(x, y, z) = x^2 + y^2 - z = 0 \implies \nabla F = 2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k} = 2\mathbf{i} + 4\mathbf{j} - \mathbf{k}$ at $(1, 2, 5)$. Tangent plane equation: $2(x - 1) + 4(y - 2) - 1(z - 5) = 0 \implies \mathbf{2x + 4y - z = 5}$.
**Practice Q2. Find and classify critical points of $f(x, y) = x^3 + y^3 - 3xy$.**
*Solution*: $f_x = 3x^2 - 3y = 0 \implies y = x^2$. $f_y = 3y^2 - 3x = 0 \implies x = y^2 \implies x = (x^2)^2 = x^4 \implies x(x^3 - 1) = 0$. Critical points: $(0, 0)$ and $(1, 1)$. Second partials: $A = f_{xx} = 6x, B = f_{xy} = -3, C = f_{yy} = 6y$. - At $(0, 0)$: $\Delta = AC - B^2 = (0)(0) - (-3)^2 = -9 < 0 \implies$ **Saddle Point**. - At $(1, 1)$: $\Delta = (6)(6) - (-3)^2 = 27 > 0$ and $A = 6 > 0 \implies$ **Relative Minimum** value $f(1, 1) = -1$.

## Topic 2.1: Convergence Tests & Series Summations

### ORIGINAL EXAM PYQ / ASSIGNMENT: Finite Series Summation (SEE 2024, 2025)

**1. Evaluate $\sum_{i=1}^{4}(2n + 3)$.**
*Solution*: Expand for $n = 1, 2, 3, 4$:

$$\sum_{i=1}^{4}(2n + 3) = (2(1) + 3) + (2(2) + 3) + (2(3) + 3) + (2(4) + 3) = 5 + 7 + 9 + 11 = \mathbf{32}$$

### EXTRA PRACTICE QUESTION: D'Alembert Ratio Test & Comparison Test

**Practice Q1. Test the convergence of the series $\sum_{n=1}^{\infty} \frac{n}{2^n}$ using Ratio Test.**
*Solution*: $a_n = \frac{n}{2^n}$, $a_{n+1} = \frac{n+1}{2^{n+1}}$.

$$L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{n+1}{2^{n+1}} \cdot \frac{2^n}{n} = \lim_{n\to\infty} \frac{n+1}{2n} = \frac{1}{2}$$

Since $L = \frac{1}{2} < 1$, the series **converges absolutely**.
**Practice Q2. Test the convergence of $\sum_{n=1}^{\infty} \frac{1}{n^2+5}$.**
*Solution*: Compare with $b_n = \frac{1}{n^2}$ (convergent $p$-series, $p = 2 > 1$).

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{n^2}{n^2 + 5} = 1 > 0 \implies \text{By Limit Comparison Test, the series } \mathbf{converges}.$$

## Topic 2.2: Taylor & Maclaurin Series Expansions

### ORIGINAL EXAM PYQ / ASSIGNMENT: Maclaurin Series Derivation for Exponential Function (Mid-Sem 2026)

**1. Determine the Taylor/Maclaurin series at $x = 0$ for $f(x) = e^x$.**
*Solution*: Derivatives: $f(x) = e^x \implies f^{(n)}(0) = 1$ for all $n \geq 0$.

$$e^x = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \mathbf{1} + \mathbf{x} + \frac{\mathbf{x^2}}{\mathbf{2!}} + \frac{\mathbf{x^3}}{\mathbf{3!}} + \cdots + \frac{\mathbf{x^n}}{\mathbf{n!}} + \ldots$$

### EXTRA PRACTICE QUESTION: Trigonometric & Logarithmic Maclaurin Expansions

**Practice Q1. Derive the Maclaurin series expansion for $f(x) = \sin x$ up to $x^5$.**
*Solution*: $f(0) = 0, f'(0) = 1, f''(0) = 0, f'''(0) = -1, f^{(4)}(0) = 0, f^{(5)}(0) = 1$.

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots = \mathbf{x} - \frac{\mathbf{x^3}}{\mathbf{6}} + \frac{\mathbf{x^5}}{\mathbf{120}} - \ldots$$

**Practice Q2. Expand $f(x) = \ln(1 + x)$ in a Maclaurin series.**
*Solution*: $f(0) = 0, f'(0) = 1, f''(0) = -1, f'''(0) = 2, f^{(4)}(0) = -6$.

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots = \sum_{\mathbf{n=1}}^{\infty} \frac{\mathbf{(-1)^{n-1} x^n}}{\mathbf{n}}$$

## Topic 3.1: Improper Integrals (Type 1 & Type 2)

### ORIGINAL EXAM PYQ / ASSIGNMENT: Type 1 Infinite Limit Evaluation (Syllabus Core Topic)

**1. Evaluate the improper integral $\int_1^{\infty} \frac{dx}{x^2}$.**
*Solution*: $\lim_{t\to\infty} \int_1^t x^{-2} dx = \lim_{t\to\infty} \left[-\frac{1}{x}\right]_1^t = \lim_{t\to\infty} \left(-\frac{1}{t} + 1\right) = 0 + 1 = \mathbf{1}$ (Convergent).

### EXTRA PRACTICE QUESTION: Type 1 & Type 2 Discontinuous Integrals

**Practice Q1. Evaluate improper integral $\int_0^{\infty} e^{-3x} dx$.**
*Solution*: $\lim_{t\to\infty} \left[-\frac{1}{3} e^{-3x}\right]_0^t = \lim_{t\to\infty} \left(-\frac{1}{3} e^{-3t} + \frac{1}{3}\right) = 0 + \frac{1}{3} = \mathbf{\frac{1}{3}}$.
**Practice Q2. Evaluate Type 2 improper integral $\int_0^1 \frac{dx}{\sqrt{x}}$.**
*Solution*: Discontinuity at $x = 0$. $\lim_{t\to 0^+} \int_t^1 x^{-1/2} dx = \lim_{t\to 0^+} [2\sqrt{x}]_t^1 = \lim_{t\to 0^+} (2 - 2\sqrt{t}) = \mathbf{2}$.

## Topic 3.2: Beta & Gamma Special Functions

### ORIGINAL EXAM PYQ / ASSIGNMENT: Beta Symmetry, Half-Integers & Negative Gamma (Re-Exam 2026)

**1. Prove that $\beta(m,n) = \beta(n,m)$.**
*Solution*: Definition $\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}dx$. Substitute $t = 1-x \implies x = 1-t, dx = -dt$.

$$\beta(m,n) = \int_1^0 (1-t)^{m-1}t^{n-1}(-dt) = \int_0^1 t^{n-1}(1-t)^{m-1}dt = \beta(\mathbf{n},\mathbf{m})$$

**2. Given $\Gamma(1.8) = 0.9314$, find $\Gamma(-2.2)$.**
*Solution*: Use recurrence formula $\Gamma(n) = \frac{\Gamma(n+1)}{n}$: 1. $\Gamma(-0.2) = \frac{\Gamma(0.8)}{-0.2}$. Note $\Gamma(1.8) = 0.8\Gamma(0.8) \implies \Gamma(0.8) = \frac{0.9314}{0.8} = 1.16425$. 2. $\Gamma(-0.2) = \frac{1.16425}{-0.2} = -5.82125$. 3. $\Gamma(-1.2) = \frac{\Gamma(-0.2)}{-1.2} = \frac{-5.82125}{-1.2} = 4.85104$. 4. $\Gamma(-2.2) = \frac{\Gamma(-1.2)}{-2.2} = \frac{4.85104}{-2.2} = \mathbf{-2.205}$.
**3. Evaluate $\Gamma(5/2)$.**
*Solution*: Apply $\Gamma(n+1) = n\Gamma(n)$ with $\Gamma(1/2) = \sqrt{\pi}$:

$$\Gamma(5/2) = \frac{3}{2}\Gamma(3/2) = \frac{3}{2}\cdot\frac{1}{2}\Gamma(1/2) = \frac{\mathbf{3\sqrt{\pi}}}{\mathbf{4}}$$

### EXTRA PRACTICE QUESTION: Beta-Gamma Integrals & Transformation Identities

**Practice Q1. Evaluate $\int_0^{\pi/2} \sin^5\theta\cos^3\theta d\theta$ using Beta-Gamma functions.**
*Solution*: Formula $\int_0^{\pi/2}\sin^p\theta\cos^q\theta d\theta = \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\Gamma\left(\frac{p+q+2}{2}\right)}$. For $p=5, q=3$: $\frac{\Gamma(3)\Gamma(2)}{2\Gamma(5)} = \frac{(2!)(1!)}{2(4!)} = \frac{2}{2(24)} = \mathbf{\frac{1}{24}}$.
**Practice Q2. Prove $\beta(m,n) = \int_0^\infty \frac{y^{n-1}}{(1+y)^{m+n}}dy$.**
*Solution*: Substitute $x = \frac{y}{1+y} \implies 1-x = \frac{1}{1+y}, dx = \frac{dy}{(1+y)^2}$.

$$\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}dx = \int_0^\infty \frac{y^{m-1}}{(1+y)^{m+n}}dy = \int_{\mathbf{0}}^{\infty} \frac{\mathbf{y^{n-1}}}{\mathbf{(1+y)^{m+n}}}\mathbf{dy} \quad \text{(by symmetry)}$$

## Topic 3.3: Bounded Area & Volume of Solids of Revolution

### ORIGINAL EXAM PYQ / ASSIGNMENT: Solids of Revolution & 2D Bounded Area (SEE 2024–25, Re-Exam 2026, Mid-Sem 2026)

**1. Volume of solid generated by revolving $y = x^2$ over $[1,3]$ about $y$-axis.**
*Solution*: Cylindrical Shell Method: $V = 2\pi\int_1^3 xydx = 2\pi\int_1^3 x(x^2)dx = 2\pi\left[\frac{x^4}{4}\right]_1^3 = 2\pi\left(\frac{81-1}{4}\right) = \mathbf{40\pi}$.
**2. Find area bounded by $y = x^2$, $x$-axis, $x = -2$, and $x = 3$.**
*Solution*: Area $= \int_{-2}^3 x^2dx = \left[\frac{x^3}{3}\right]_{-2}^3 = \frac{27-(-8)}{3} = \mathbf{\frac{35}{3}}$.

### EXTRA PRACTICE QUESTION: Disk Method Volume & Surface Area of Revolution

**Practice Q1. Find volume generated by revolving $y = \sqrt{x}$ over $[0,4]$ about $x$-axis.**
*Solution*: Disk Method: $V = \pi\int_0^4 y^2dx = \pi\int_0^4(\sqrt{x})^2dx = \pi\left[\frac{x^2}{2}\right]_0^4 = \mathbf{8\pi}$.
**Practice Q2. Find surface area generated by revolving $y = 2\sqrt{x}$ over $[0,3]$ about $x$-axis.**
*Solution*: $S = 2\pi\int y\sqrt{1+(y')^2}dx$. $y' = \frac{1}{\sqrt{x}} \implies \sqrt{1+(y')^2} = \sqrt{\frac{x+1}{x}}$.

$$S = 2\pi\int_0^3 2\sqrt{x}\sqrt{\frac{x+1}{x}}dx = 4\pi\int_0^3\sqrt{x+1}dx = 4\pi\left[\frac{2}{3}(x+1)^{3/2}\right]_0^3 = \frac{8\pi}{3}(8-1) = \mathbf{\frac{56\pi}{3}}$$

## Topic 4.1: Double & Triple Integrals (Cartesian & Polar)

### ORIGINAL EXAM PYQ / ASSIGNMENT: Iterated 3D Triple & 2D Double Integrals (SEE 2024, 2025)

**1. Evaluate 3D Triple Integral $\int_0^2 \int_1^3 \int_1^2 xy^2 z \, dz \, dy \, dx$.**
*Solution*: Product of independent integrals:

$$\int_0^2 x dx = \left[\frac{x^2}{2}\right]_0^2 = 2, \quad \int_1^3 y^2 dy = \left[\frac{y^3}{3}\right]_1^3 = \frac{26}{3}, \quad \int_1^2 z dz = \left[\frac{z^2}{2}\right]_1^2 = \frac{3}{2}$$

$$\text{Total} = 2 \cdot \frac{26}{3} \cdot \frac{3}{2} = \mathbf{26}$$

**2. Evaluate $\int_0^1 \int_0^2 y^2 x \, dy \, dx$.**
*Solution*: $\int_0^1 x \left[\frac{y^3}{3}\right]_0^2 dx = \frac{8}{3} \int_0^1 x dx = \frac{8}{3} \cdot \frac{1}{2} = \mathbf{\frac{4}{3}}$.

### EXTRA PRACTICE QUESTION: Polar Double Integrals & Changing Order of Integration

**Practice Q1. Evaluate $\int_0^\pi \int_0^1 r^3 \sin\theta \, dr \, d\theta$.**
*Solution*: $\left(\int_0^1 r^3 dr\right)\left(\int_0^\pi \sin\theta d\theta\right) = \left[\frac{r^4}{4}\right]_0^1 [-\cos\theta]_0^\pi = \left(\frac{1}{4}\right)(1-(-1)) = \mathbf{\frac{1}{2}}$.
**Practice Q2. Evaluate by changing order of integration: $\int_0^1 \int_y^1 e^{x^2} dx dy$.**
*Solution*: Region $0 \le y \le 1, y \le x \le 1 \implies 0 \le x \le 1, 0 \le y \le x$.

$$\int_0^1 \left(\int_0^x e^{x^2} dy\right) dx = \int_0^1 x e^{x^2} dx = \left[\frac{1}{2} e^{x^2}\right]_0^1 = \frac{\mathbf{e-1}}{\mathbf{2}}$$

## Topic 4.2: Enclosed Area Applications by Double Integration

### ORIGINAL EXAM PYQ / ASSIGNMENT: Ellipse, Parabola-Line & Polar Circles Area (SEE 2024, 2025, Re-Exam 2026)

**1. Total area of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ by double integration.**
*Solution*: Area $= 4 \int_0^a \int_0^{b\sqrt{1-x^2/a^2}} dy dx = 4b \int_0^a \sqrt{1-x^2/a^2} dx = 4b\left(\frac{a\pi}{4}\right) = \pi\mathbf{ab}$.
**2. Area between parabola $y = 4x - x^2$ and line $y = x$.**
*Solution*: Intersections $4x - x^2 = x \implies x(3-x) = 0 \implies x = 0, 3$.

$$\text{Area} = \int_0^3 (4x - x^2 - x) dx = \int_0^3 (3x - x^2) dx = \left[\frac{3x^2}{2} - \frac{x^3}{3}\right]_0^3 = \frac{27}{2} - 9 = \mathbf{\frac{9}{2}}$$

**3. Polar area bounded by $r = 2\sin\theta$ and $r = 4\sin\theta$.**
*Solution*: Area between concentric polar circles: Area $= \frac{1}{2}\int_0^\pi (r_2^2 - r_1^2) d\theta = \frac{1}{2}\int_0^\pi (16\sin^2\theta - 4\sin^2\theta) d\theta = 6\int_0^\pi \sin^2\theta d\theta = 6\left(\frac{\pi}{2}\right) = \mathbf{3}\pi$.

### EXTRA PRACTICE QUESTION: Cardioid Area & Parabolic Bounded Regions

**Practice Q1. Find the total area enclosed by cardioid $r = a(1+\cos\theta)$.**
*Solution*: Area $= 2 \cdot \frac{1}{2}\int_0^\pi r^2 d\theta = a^2 \int_0^\pi (1 + 2\cos\theta + \cos^2\theta) d\theta = a^2(\pi + 0 + \pi/2) = \mathbf{\frac{3}{2}}\pi\mathbf{a^2}$.
**Practice Q2. Find the area of the region bounded by $y = x^2$ and $y = 2x$.**
*Solution*: Intersections $x^2 = 2x \implies x = 0, 2$. Area $= \int_0^2 (2x - x^2) dx = \left[x^2 - \frac{x^3}{3}\right]_0^2 = 4 - \frac{8}{3} = \mathbf{\frac{4}{3}}$.

## Topic 4.3: Vector Calculus (Divergence, Curl & Vector Theorems)

### ORIGINAL EXAM PYQ / ASSIGNMENT: Divergence Evaluation & Classical Vector Theorems (Re-Exam 2026)

**1. For $\mathbf{A} = x^2z\mathbf{i} - 2y^3z^2\mathbf{j} + xy^2z\mathbf{k}$, find $\nabla \cdot \mathbf{A}$ at $(1, -1, 1)$.**
*Solution*: $\nabla \cdot \mathbf{A} = \frac{\partial}{\partial x}(x^2z) + \frac{\partial}{\partial y}(-2y^3z^2) + \frac{\partial}{\partial z}(xy^2z) = 2xz - 6y^2z^2 + xy^2$. At $(1, -1, 1)$: $2(1)(1) - 6(1)(1) + (1)(1) = 2 - 6 + 1 = \mathbf{-3}$.

### EXTRA PRACTICE QUESTION: Irrotational Vector Fields & Solenoidal Constants

**Practice Q1. Show that $\mathbf{F} = (y^2\cos x + z^3)\mathbf{i} + (2y\sin x - 4)\mathbf{j} + (3xz^2 + 2)\mathbf{k}$ is irrotational.**
*Solution*: Compute $\nabla \times \mathbf{F} = \mathbf{i}(0 - 0) - \mathbf{j}(3z^2 - 3z^2) + \mathbf{k}(2y\cos x - 2y\cos x) = \mathbf{0} \implies$ Irrotational.
**Practice Q2. Find constant $a$ such that $\mathbf{V} = (x + 3y)\mathbf{i} + (y - 2z)\mathbf{j} + (x + az)\mathbf{k}$ is solenoidal.**
*Solution*: Solenoidal condition $\nabla \cdot \mathbf{V} = 0 \implies 1 + 1 + a = 0 \implies \mathbf{a = -2}$.

## Topic 5.1: Ordinary & Partial Differential Equations

### ORIGINAL EXAM PYQ / ASSIGNMENT: 1st Order LDE, Cauchy-Euler & Lagrange PDEs (Syllabus Core Topics)

**1. Solve 1st-Order LDE $\frac{dy}{dx} + P(x)y = Q(x)$.**
*Solution*: Integrating Factor IF $= e^{\int P dx}$. General solution: $y \cdot (\text{IF}) = \int Q(x)(\text{IF})dx + C$.
**2. Solve Cauchy-Euler Equation $x^2\frac{d^2y}{dx^2} + px\frac{dy}{dx} + qy = 0$.**
*Solution*: Substitute $x = e^z \implies z = \ln x$. Replace $x\frac{dy}{dx} = D_z y$ and $x^2\frac{d^2y}{dx^2} = D_z(D_z - 1)y$.

### EXTRA PRACTICE QUESTION: ODE Integrating Factors & Cauchy-Euler Practice

**Practice Q1. Solve $\frac{dy}{dx} + 2xy = 4x$.**
*Solution*: IF $= e^{\int 2xdx} = e^{x^2}$. Solution: $ye^{x^2} = \int 4xe^{x^2}dx + C = 2e^{x^2} + C \implies \mathbf{y = 2 + Ce^{-x^2}}$.
**Practice Q2. Solve $x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 4y = 0$.**
*Solution*: $x = e^z \implies (D_z^2 - 4D_z + 4)y = 0 \implies (m - 2)^2 = 0 \implies y = (C_1 + C_2 z)e^{2z} = \mathbf{(C_1 + C_2 \ln x)x^2}$.
**Practice Q3. Solve 1st-order PDE $y^2p - xyq = x(z - 2y)$.**
*Solution*: Lagrange auxiliary: $\frac{dx}{y^2} = \frac{dy}{-xy} = \frac{dz}{x(z-2y)} \implies x^2 + y^2 = C_1$ and $yz - y^2 = C_2 \implies \mathbf{\Phi(x^2 + y^2, yz - y^2) = 0}$.

### CASIO fx-991ES PLUS PRO-TIP: CASIO fx-991ES PLUS Button Sequences & Exam Blueprint

- **Limits ('CALC' mode)**: Type expression → Press `CALC` → Enter $X = 5.0001$ or $0.0001$ → Press `=`.
- **Numerical Derivatives ('d/dx')**: Press `SHIFT` + `dx` → Type $f(X)$ → Enter $x = x_0$ → Press `=`.
- **3D Vector Dot Product**: Press `MODE 8 (VECTOR)` → Enter `VctA` and `VctB` → `SHIFT 5` → `VctA · VctB`.
- **60-Mark SEE Strategy**: 90 mins for Section C (30 Marks), 50 mins for Section B (20 Marks), 25 mins for Section A (10 Marks).